

# Butterfly scan

The algorithms and programs described here are in the repository on GitHub:

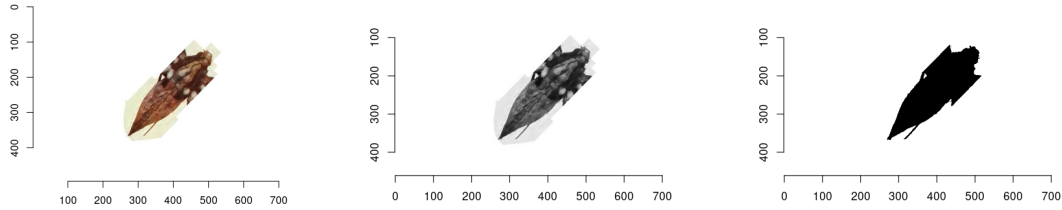
[https://github.com/andy-aa/butterfly\\_scan](https://github.com/andy-aa/butterfly_scan)

The algorithms are implemented in the R programming language[1].

## Calculation of the body volume of a butterfly

The algorithm for estimating the volume of the body of a butterfly:

1. Converting a color image to black and white. Three-channel RGB image  $\rightarrow$  Single-channel grayscale image  $\rightarrow$  Single-channel black and white images. When converting an image to black and white, it is necessary to use an empirical coefficient that is in the range from 0 to 1. The empirical coefficient is a color code used to separate all colors in an image into two groups, black and white.



2. Search for the axis of rotation of an arbitrarily oriented body of a butterfly. The axis of rotation is a line connecting the two most distant points from each other. To reduce the search time, the convex hull of the figure is first constructed.

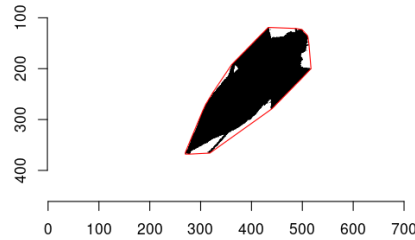


Figure 1: Convex hull

3. In the set of points of the convex hull, we need to find two points that have a maximum distance function:  $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ .

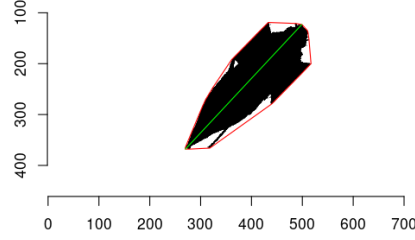
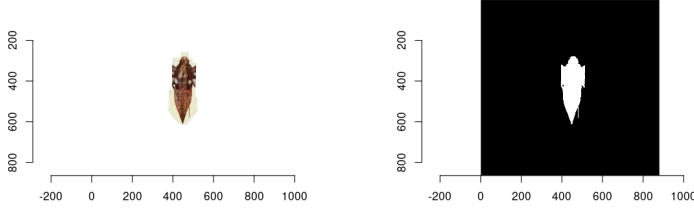


Figure 2: Axis of rotation

4. Finding the angle between the rotation axis and the x-axis. Rotate the image so that the rotation axis is perpendicular to the x-axis and parallel to the y-axis.

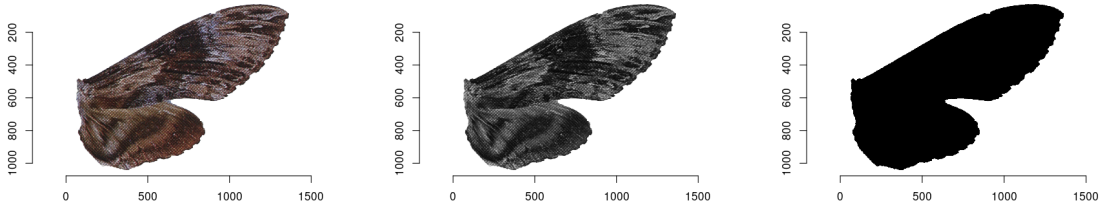


5. Dividing the silhouette (contour) into layers parallel to the x-axis. Finding the width of each layer. The width ( $w_i$ ) of a layer is equal to the number of pixels in that layer multiplied by the physical size of a pixel. The layer height ( $h$ ) is equal to the physical pixel size. Layer volume is:  $v_i = (w_i/2)^2 * \pi * h$ . The total volume ( $V$ ) is the sum of the volumes of its layers  $V = \sum_{i=1}^n v_i$ .

## Calculation of the moment of inertia of a butterfly wing

The algorithm for estimating the moment of inertia of a butterfly wing:

1. Converting a color image to black and white.

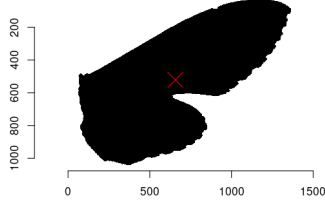


2. Calculation of the coordinates of the conditional center of mass. The classic formula for calculating the

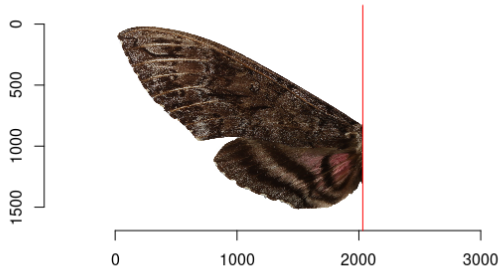
center of mass is:  $\vec{r}_c = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$ . In the case of a two-dimensional object, the formula takes the form:

$$x_c = \frac{\sum_{i=1}^n m_i x_i}{\sum_{i=1}^n m_i} \text{ and } y_c = \frac{\sum_{i=1}^n m_i y_i}{\sum_{i=1}^n m_i}; \text{ If we take the mass of each point equal to 1, the formulas are}$$

noticeably simplified:  $x_c = \bar{x}$  and  $y_c = \bar{y}$ .



3. The classical formula for calculating the **moment of inertia** is:  $J_p = \sum_{i=1}^n m_i r_i^2$ , where  $m_i$  is the mass of  $i$ -th point;  $r_i$  is the distance from the  $i$ -th point to the axis. If we take the mass of each point equal to 1, and for the point of the axis we take the center of mass, the formula for **polar moment of inertia** will take the form:  $J_p = \sum_{i=1}^n r_i^2 = \sum_{i=1}^n ((x_i - x_c)^2 + (y_i - y_c)^2)$
4. In our case **moment of inertia about the axis** will be calculated relative to the wing attachment line. This is due to the way the wing is attached to the body and how it moves during flight [2]. The formula for calculating the moment of inertia about the axis will take the form:  $J_a = \sum_{i=1}^n r_i^2 = \sum_{i=1}^n (x_i - x_a)^2$ , where  $x_a$  is the distance to the axis of rotation.



## Geometric classification and measurement of wing parameters

In order to highlight the characteristic features of the butterfly wings, it is of interest to approximate the butterfly wing using polygons[3]. A simplified representation of a wings shape allows you to highlight the main geometric properties.

Left wing of *Hemaris diffinis*

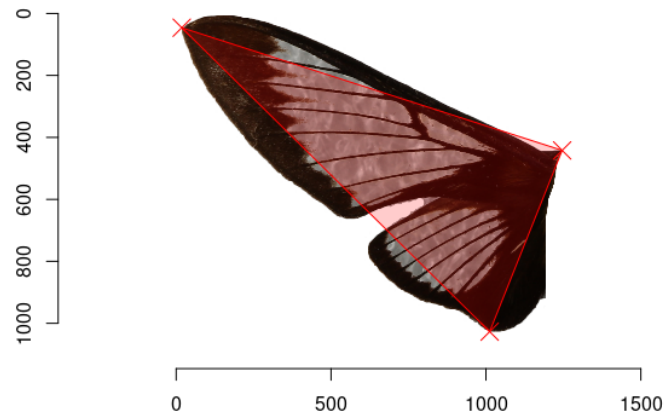


Figure 3: Threangle

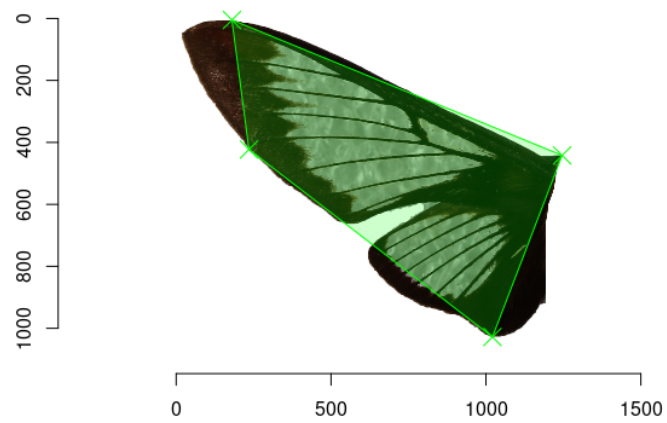


Figure 4: Quadrangle

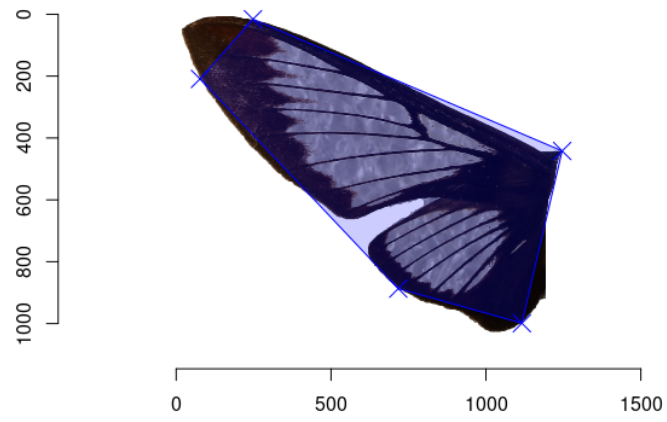


Figure 5: Pentagon

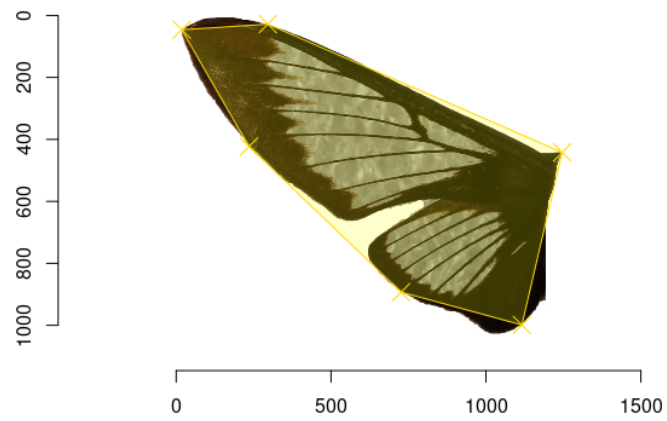


Figure 6: Hexagon

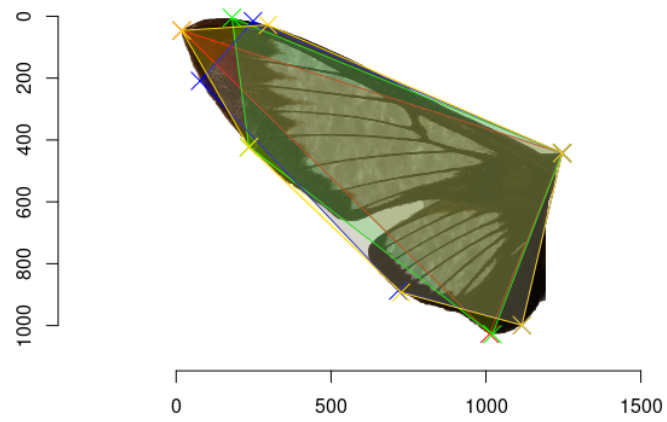


Figure 7: Superposition of polygons

Left wing of *Manduca rustica*

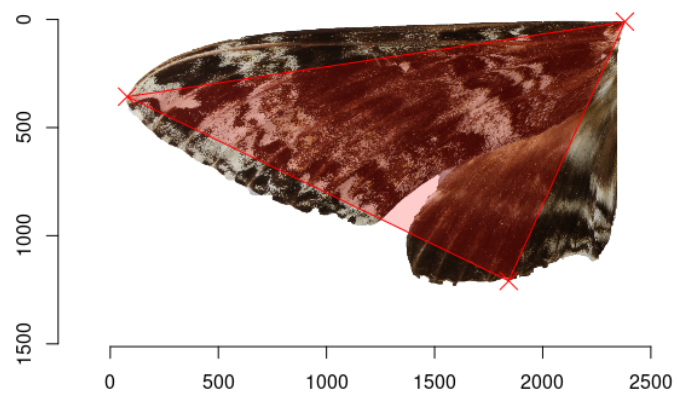


Figure 8: Threangle

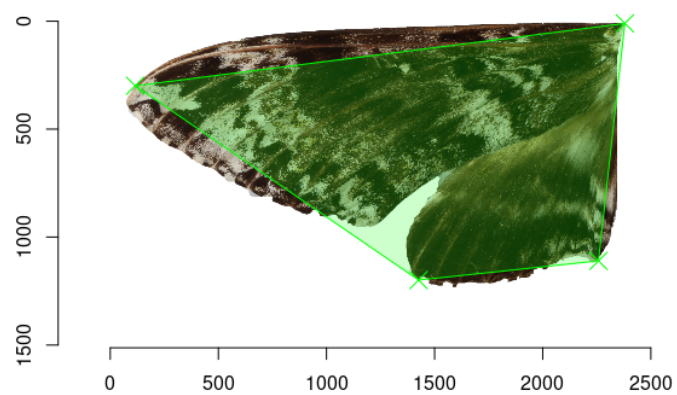


Figure 9: Quadrangle

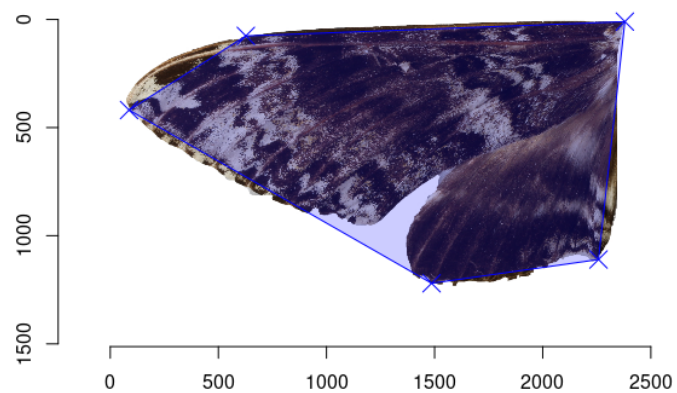


Figure 10: Pentagon

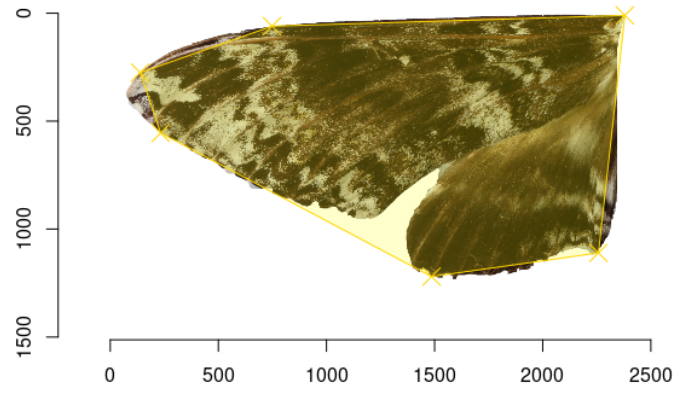


Figure 11: Hexagon

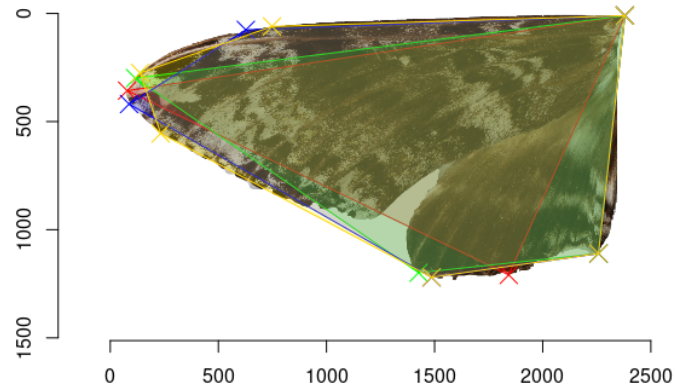


Figure 12: Superposition of polygons

### Butterfly wing boundary definition

To find the boundaries of objects in images, one of the most common algorithms is the Canny algorithm [4].



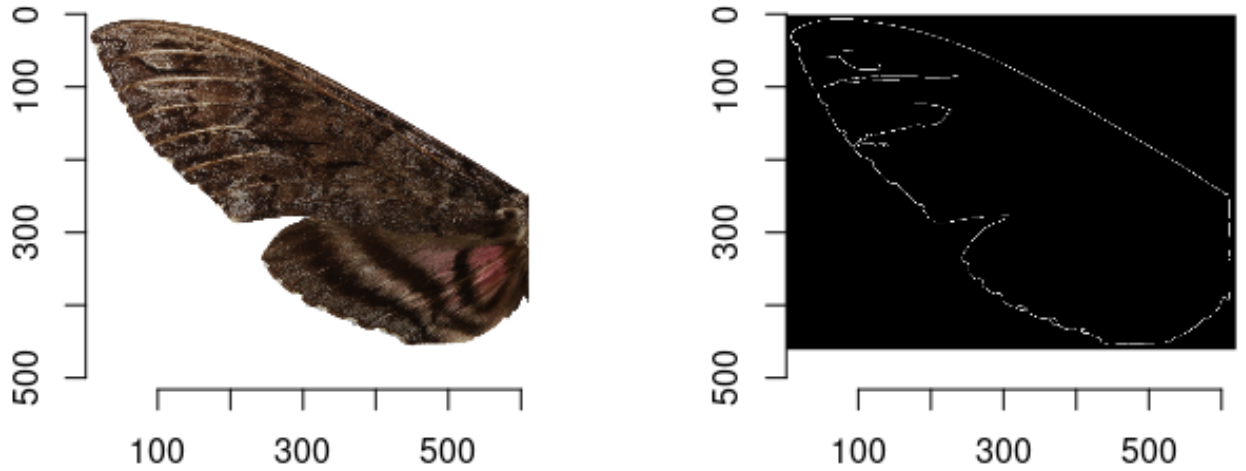


Figure 13: Finding edges in a color image

When using this algorithm, the boundaries will be found not only outside the object but also inside. To avoid this, we can first convert the image to black and white. The disadvantage of using this algorithm is that at the output we have an unordered point cloud, which is the edge of the object.

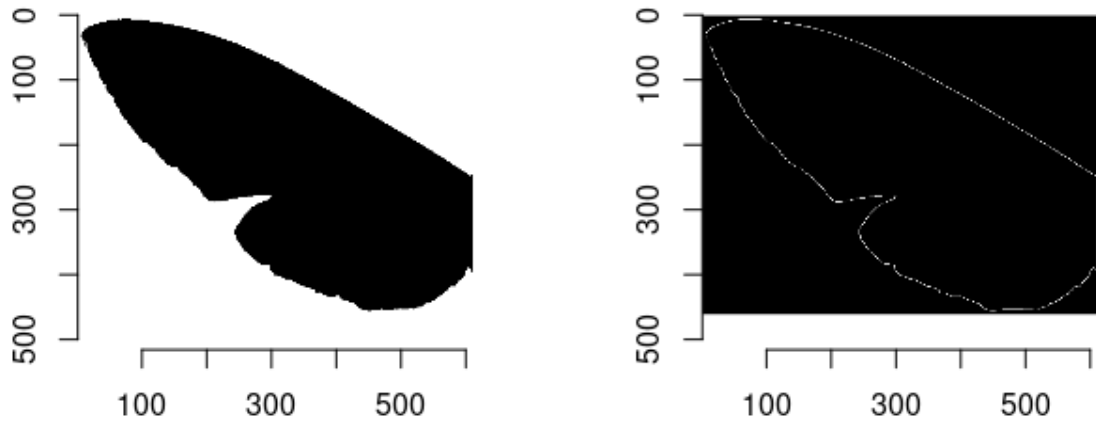


Figure 14: Finding edges in a contrasting black and white image

The boundary of the butterfly wing object is actually a concave hull. One of the algorithms for finding the edges of a concave hull is the Alpha shape algorithm [5].

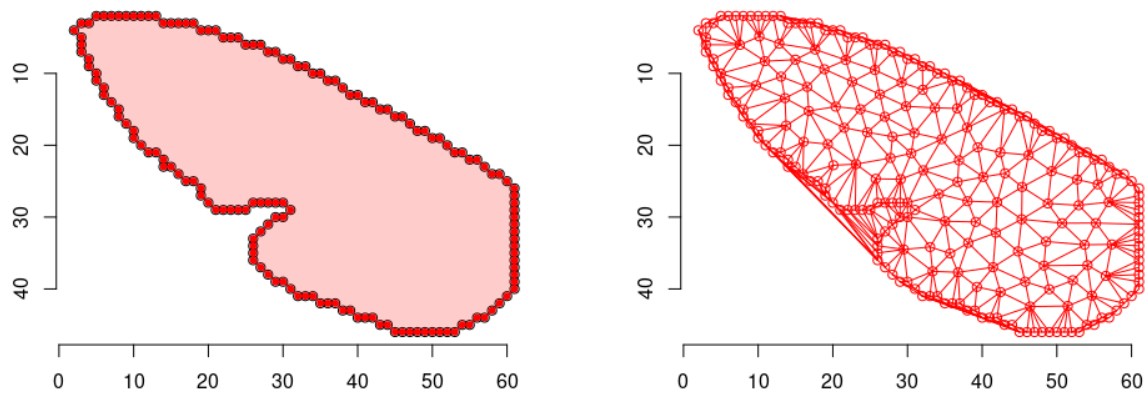


Figure 15: Concave Hull of a wing object

As a result of the Alpha shape algorithm, we can get a polygon that does not intersect itself and has no holes. To find the area of such polygon, we can use the shoelace Gauss algorithm. [6]

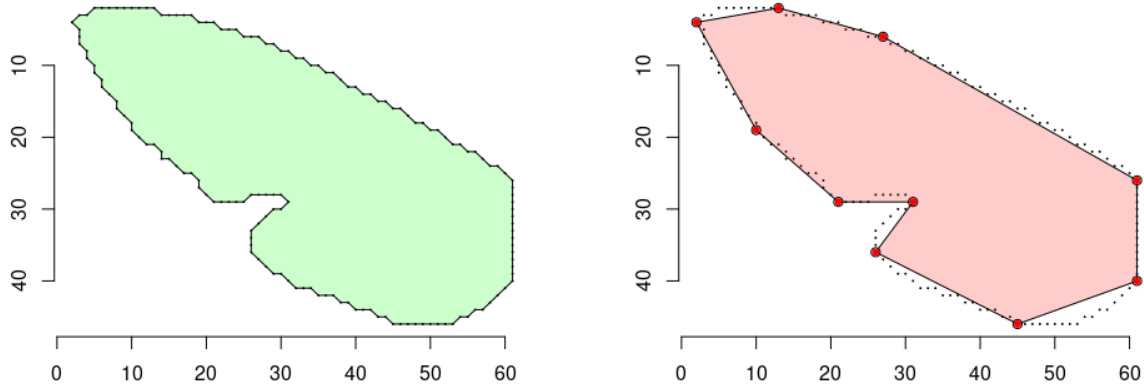


Figure 16: Concave hull after minimizing the number of points

## Running the program for execution

Image files should be located as shown in the following figure.

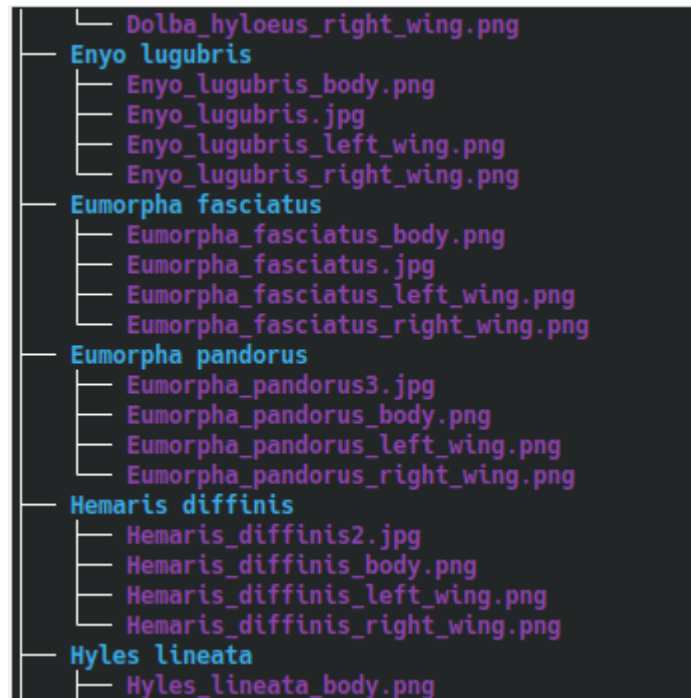
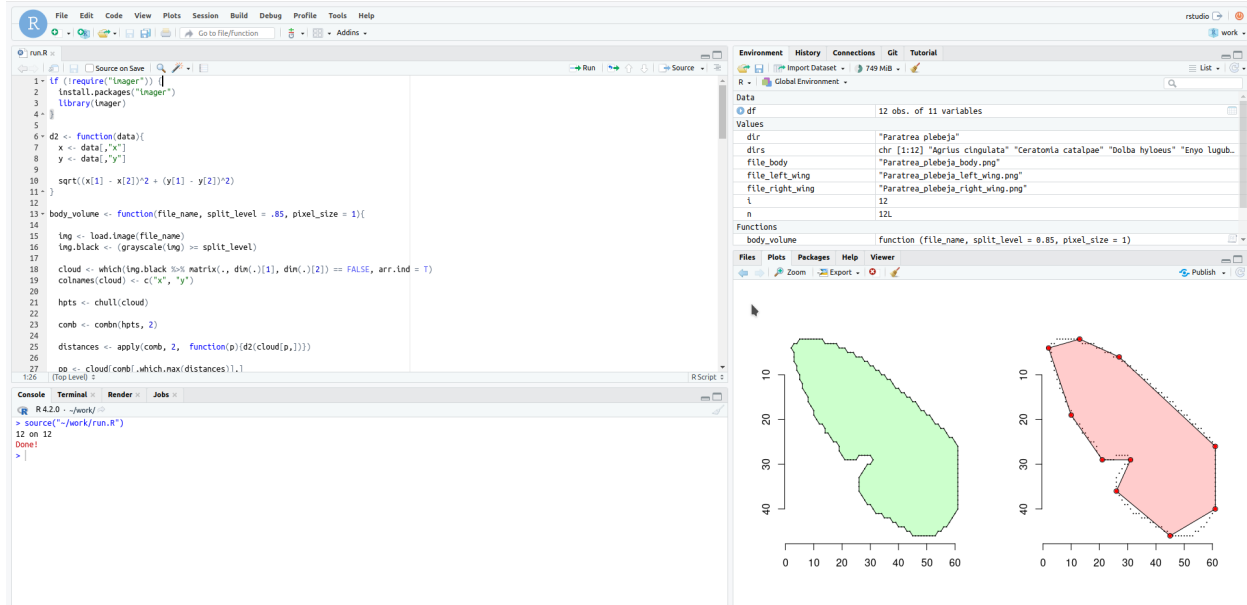


Figure 17: Folder tree

For each individual moth, a folder should be created with the same name as the name of the insect. This name will later be inserted into the results table. Inside the folder there must be files whose names can be anything, but at the end of the file name there must be postfixes: **body**, **left\_wing** or **right\_wing**.

To run the program for execution, you need to open the file in the **Rstudio** program and execute the command **Code** → **Source**



After execution, a file called **data.csv** will appear in the folder.

Import Text Data

File/URL:  Update

Data Preview:

| Name                     | body_volume | body_area | body_height | body_width | right_wing_inertia_polar | right_wing_inertia_axis | right_wing_area | left_wing_inertia_polar | left_wing_inertia_axis | left_wing_area |
|--------------------------|-------------|-----------|-------------|------------|--------------------------|-------------------------|-----------------|-------------------------|------------------------|----------------|
| (character)              | (double)    | (double)  | (double)    | (double)   | (double)                 | (double)                | (double)        | (double)                | (double)               | (double)       |
| Agrius cingulata         | 1336036.1   | 850547    | 2138        | 524        | 593038493964             | 1.848049e+12            | 3246292         | 636138446574            | 1.781543e+12           | 3246292        |
| Ceratomia catalpae       | 618802.6    | 393942    | 1388        | 403        | 237741645532             | 6.073490e+11            | 2097624         | 238878353367            | 6.372202e+11           | 2097624        |
| Dolba hyloeus            | 964533.3    | 614041    | 1734        | 486        | 278863325553             | 6.985548e+11            | 2286588         | 29619895888             | 7.056534e+11           | 2286588        |
| Enyo lugubris            | 1227927.6   | 781723    | 1837        | 558        | 249463956892             | 6.460610e+11            | 2062710         | 245735043941            | 6.443143e+11           | 2062710        |
| Eumorphia fasciatus      | 1432922.8   | 912227    | 2356        | 562        | 466617996202             | 1.240496e+12            | 2830082         | 495024241344            | 1.254596e+12           | 2830082        |
| Eumorphia pandorus       | 1284740.2   | 817891    | 1949        | 562        | 479104450703             | 1.411579e+12            | 2922756         | 472147529373            | 1.303395e+12           | 2922756        |
| Hemeris diffrinis        | 992578.3    | 631895    | 1606        | 489        | 85684420584              | 2.174119e+11            | 1162754         | 94563784964             | 2.455001e+11           | 1162754        |
| Hyles lineata            | 1764491.2   | 1123310   | 2183        | 703        | 750458641112             | 2.267393e+12            | 3667382         | 756435291823            | 2.264223e+12           | 3667382        |
| Manduca quinquemaculatus | 1174280.2   | 747570    | 1896        | 547        | 767779738642             | 2.133289e+12            | 3681822         | 810732338025            | 1.900974e+12           | 3681822        |
| Manduca rustica          | 1580532.1   | 1006198   | 2074        | 620        | 883888541371             | 2.626349e+12            | 4010730         | 853333396483            | 2.536054e+12           | 4010730        |
| Manduca sexta            | 1191288.8   | 758398    | 1884        | 511        | 911420046355             | 2.403966e+12            | 3973578         | 936894489341            | 2.393423e+12           | 3973578        |
| Paratrea plebeja         | 1042927.1   | 663948    | 1867        | 529        | 277913253545             | 8.296795e+11            | 2244116         | 28759552808             | 7.755945e+11           | 2244116        |

Import Options:

Name:  First Row as Names ☒ Delimiter:  Escape:  Skip:  Trim Spaces ☒ Quotes:  Local:  Open Data Viewer ☒ Comment:  NA:

Code Preview:

```
library(readr)
data <- read_csv("data.csv")
View(data)
```

Import Cancel

Figure 18: The results of the calculations

To run the program on *Palmetto* cluster for high performance, you need to execute the the following commands.

Copy files to palmetto:

```
$ git clone https://github.com/andy-aa/butterfly_scan.git
```

Make a reservation computing node:

```
$ qsub -I -l select=1:ncpus=4:mem=10gb:interconnect=1g,walltime=2:00:00
```

Load the R language module:

```
$ module load r/4.1.2-gcc/8.4.1
```

Run script for execution:

```
$ Rscript run.R
```

## References

1. R Core Team (2022) R: A language and environment for statistical computing, Vienna, Austria, R Foundation for Statistical Computing.
2. Smith DS (1965) Flight muscles of insects. *Scientific American* 212: 76–88.
3. Bronstein EM (2008) Approximation of convex sets by polytopes. *J Math Sci* 153: pages727–762.
4. Canny J (1986) A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 8: 679–698.
5. Knorr EM, Ng RT, Shilvock DL (1997) Finding boundary shape matching relationships in spatial data, In: Scholl M, Voisard A (Eds.), *Advances in spatial databases*, Berlin, Heidelberg, Springer Berlin Heidelberg, 29–46.
6. Allgower PH Eugene L.; Schmidt (1986) Computing volumes of polyhedra. *Mathematics of Computation* 46: 171–174.